

# Damián Pablo San Román Alerigi, PhD

## Extended Curriculum Vitae — Full Research, Patent & Career Record

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## COMPANION DOCUMENT

This Extended CV accompanies the two-page résumé and provides the full publication, patent and career record behind it. Portfolio total: 8 peer-reviewed journal articles, 4 industry/technical journal articles, ~50 conference papers, 2 preprints, 1 doctoral thesis, 26 granted patents (11 pending), and 444 citations (h-index 11, i10-index 10; Google Scholar, September 2026).

## CAREER ARC

**Phase I — Theoretical & computational photonics (KAUST, 2009–2014).** Founding cohort member of the KAUST Photonics Master's programme; M.S. (2010) and PhD (2014) in Electrical Engineering under Prof. Boon S. Ooi. Work centred on light propagation in heterogeneous, nonlinear and time-varying media — computational electromagnetics, beam shaping and chalcogenide-glass material engineering. Thesis: *Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light* (KAUST, 2014).

**Phase II — Applied laser–matter interaction (Saudi Aramco, 2015–2019).** Transition from theory to industrial application: numerical modelling of laser–rock and electromagnetic–rock interaction, thermal/geomechanical coupling, and the early laser-perforation programme. First-author work established the physics models underpinning the technology.

**Phase III — Technology deployment & diversification (2020–present).** Co-PI on Aramco's high-power laser programme with Sameeh I. Batarseh, carrying the technology from laboratory demonstration to first industrial field deployment. In parallel, opened three new research lines: hydrogen-sulfide valorisation and solar fuels; photonic and machine-learning sensing for subsurface and surface assets; and hollow-core optical fibre for kilowatt-class downhole power delivery.

## ACHIEVEMENT CLUSTERS

### High-power laser technology: laboratory to first industrial deployment

The defining programme of the career, documented continuously in the peer-reviewed and industry record over a decade:

- **Physics & modelling foundation (2016–2019)** — first-authored the governing numerical treatments of laser–rock interaction (ARMA-2016-142), thermal/geomechanical dynamics of electromagnetic heating (SPE-183917-MS, SPE-188790-MS), and a scalable numerical solution for subsurface wave propagation (SPE-194888-MS, sole author). The coupled thermal–mechanical model was published in *SPE Journal* (2019).
- **Technology maturation (2019–2022)** — laser perforation as a completion technology (SPE-197192-MS), performance in carbonates and organic-rich shales (SPE-197480-MS; SPE-204809-MS, first author), laser drilling in extremely hard formations (OTC-30416-MS), and laser cladding for corrosion prevention (SPE-201617-MS).
- **First field deployments (2021–2024)** — first industrial flowline-descaling deployment (Dec. 2021; SPE-209972-MS), underlying principles first-authored (SPE-209977-MS); first industrial high-power laser perforation deployment (SPE-215016-MS, 2023) and its field validation (SPE-220900-MS, 2024).
- **Extension & scale-up (2023–2025)** — laser drilling milestones (SPE-215166-MS), photonic–hydraulic hybrid fracturing lowering breaking pressure (SPE-216510-MS, SPE-215162-MS), a long-range descaling robot (SPE-222608-MS), sweet-spot creation in tight/unconventional reservoirs (SPE-227837-MS, 2025), and net-zero reframing of the programme (SPE-224865-MS, 2025).

## Hollow-core optical fibre for kilowatt-class power delivery

Co-inventor with the University of Southampton Optoelectronics Research Centre (Poletti/Richardson group) on nested anti-resonant nodeless fibre (NANF) and corrugated-cladding fibre enabling multi-kilowatt continuous-wave transmission at 1050–1100 nm over the distances required for downhole delivery — the enabling physical layer for the laser programme above (US 11,555,958 B2 + EP 4,067,952 B1; US 11,585,976 B2 + EP 4,067,735 A1 + SA 122,430,881 B1).

## Hydrogen sulfide valorisation and solar fuels

A sustainability research line combining peer-reviewed publication with a protected IP position — converting a waste stream into hydrogen rather than elemental sulfur: senior-author perspective in *Advanced Energy and Sustainability Research* (2023, with the KAUST Laquai group); UV-induced decomposition pathways in *RSC Advances* (2025, up to 52% conversion within 60 minutes); and a patent cluster spanning photocatalytic conversion (US 12,559,366 B2, lead inventor), dissociation via reactive sulfur species (US 12,459,816 B2), photolytic conversion in produced-gas streams (US 2024/0060411 A1), and steam-methane reforming (US 2025/0368507 A1).

## Photonic and machine-learning sensing

A coherent programme applying optics, spectroscopy and machine learning to characterisation and monitoring: subsurface property inference from early ML applications (SPE-195118-MS, 2019, first author) to hyperspectral multiphysics inference (SPIE, 2023, first author), protected in US 12,682,216 B2 (sole inventor) and US 12,487,075 B2; asset monitoring and computer vision (IPTC-23632-MS; US 12,430,795 B2; US 11,725,504 B2, five-jurisdiction family; the 2026 wellhead-displacement trio using beam deflection, speckle reflectometry and shadowgraphy); emissions/sustainability (US 12,394,183 B2); and chemical/fibre sensing (US 11,867,629 B2; US 12,253,459 B2; US 12,359,554 B2; US 2024/0141777 A1).

## Theoretical physics and computational electromagnetics

The scientific foundation of the career: optical analogues of gravitational attractors (*Optics Express*, 2013, first author — the most-cited work, featured in *Laser Focus World*, July 2013); Bessel–Gauss beam shaping from heterogeneous refractive-index maps (*JOSA A*, 2012, first author); electron-irradiation-induced permittivity reduction in chalcogenide glass (*J. Appl. Phys.*, 2013, first author); a high-order finite-volume method for Maxwell's equations in time-varying media (arXiv:2307.11842, first author, with D. I. Ketcheson, 2023); and a revisitation of the quantum beam splitter without a vacuum state (*Rev. Mex. Fis. E*, 2022).

## PROFESSIONAL EXPERIENCE — FULL RECORD

### Applied Physicist — Co-PI, High-Power Laser Programme

2015 – Present · Upstream Advanced Research Center, Saudi Aramco, Dhahran, KSA

Co-leads the physics, modelling and field-execution programme for high-power laser technology; see Achievement Clusters for the full technical narrative. Opened and now leads three additional research lines: hydrogen-sulfide valorisation, photonic/ML sensing, and hollow-core fibre power delivery.

### Graduate Researcher, Photonics Laboratory

2009 – 2014 · KAUST, Thuwal, KSA · Advisor: Prof. Boon S. Ooi

Founding member of the KAUST Photonics Master's programme. Co-founder and lead developer of the *Maxwell-GEMS* project (Maxwell Generalized ElectroMagnetic Solvers) — a toolbox (FEM, FDTD, spectral and paraxial methods) for wave propagation in heterogeneous, anisotropic, non-linear and time-varying media.

### Early Research Fellowships

2006 – 2009 · Mexico, United States, Spain

- **Junior Research Fellow**, Advanced Optics Laboratory, College of Sciences, UNAM (Sept. 2006 – Aug. 2009) — experiments in quantum optics; developed a measurement strategy for concurrent triple coincidences and theory on properties of vacuum states. Supervisor: Víctor M. Velázquez.
- **Research Fellow**, Joint Quantum Institute, Atomic and Molecular Optics Laboratory, University of Maryland at College Park (Mar. – Sept. 2007) — development of a laser control system for quantum-state measurement and preparation. Supervisor: Luis Orozco.
- **Research Fellow**, Institut de Ciències Fotòniques (ICFO), Spain (Mar. – Oct. 2008) — development of a fast polarimetry system and density-function measurement device for quantum-state characterisation. Supervisor: Morgan Mitchell.

## EDUCATION

| DEGREE | INSTITUTION | DATES |
|--------|-------------|-------|
|--------|-------------|-------|

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------|
| PhD, Electrical Engineering (Photonics) — Thesis: <i>Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light</i> | KAUST, Saudi Arabia                            | 2011 – 2014 |
| M.S., Electrical Engineering (Photonics)                                                                                                                                                               | KAUST, Saudi Arabia — Founding cohort          | 2009 – 2010 |
| Licenciatura (B.S.), Physics — GPA 9.67/10                                                                                                                                                             | Universidad Nacional Autónoma de México (UNAM) | 2003 – 2009 |

## COLLABORATION NETWORK

| COLLABORATOR(S)                                                 | INSTITUTION                   | DOMAIN                                                                  |
|-----------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------|
| Sameeh I. Batareseh                                             | Saudi Aramco                  | High-power laser downhole technology (principal long-term collaborator) |
| Boon S. Ooi, Tien Khee Ng                                       | KAUST Photonics Lab           | Photonics, nanostructures, computational electromagnetics               |
| Francesco Poletti, David Richardson, Hesham Sakr                | University of Southampton ORC | Hollow-core anti-resonant optical fibre                                 |
| Frédéric Laquai, Jan Kosco                                      | KAUST                         | Photocatalysis, solar fuels                                             |
| Aamir Farooq                                                    | KAUST                         | Combustion and H <sub>2</sub> S photochemistry                          |
| David I. Ketcheson                                              | KAUST                         | Numerical analysis, high-order methods                                  |
| Jennifer L. Miskimins, Ramona M. Graves                         | Colorado School of Mines      | Perforation / laser-rock interaction modelling                          |
| Adrian Cavazos Sepúlveda, Weichang Li, Yong Ma, José O. Álvarez | Saudi Aramco                  | Sensing, computer vision, chemical fingerprinting                       |

## PUBLICATIONS — JOURNAL & INDUSTRY ARTICLES (FULL RECORD)

### Peer-reviewed journal articles (8)

H.A. Khan; A. Elkhazraji; M. Abou-Daher; D.P. San Roman Alerigi; K. Hazazi; A. Farooq, "Photochemical valorization of hydrogen sulfide: a study of UV-induced decomposition pathways," **RSC Advances** **15(44)**, 37299-37308; **Royal Society of Chemistry** (10.1039/d5ra04250j)

M. Ferree; J. Kosco; F. Laquai; A. Cavazos Sepulveda; D.P. San Roman Alerigi, "Solar Fuel Production from Hydrogen Sulfide: An Upstream Energy Perspective," **Advanced Energy and Sustainability Research** **4(8)**, 2200201; **Wiley** (10.1002/aesr.202200201) **SENIOR AUTHOR**

A. Nahmad; D.P. San-Roman-Alerigi; E.M. Hernandez Gonzalez; E. Barrios; G. Armendariz Pena; V.M. Velazquez Aguilar, "The quantum beam splitter revisited without a vacuum state," **Revista Mexicana de Fisica E** **19(1)**, 010210; **Sociedad Mexicana de Fisica** (10.31349/revmexfise.19.010210)

Y. Han; Y. Fang; D.P. San Roman Alerigi; S.I. Batareseh, "Thermal/Mechanical Interaction in Laser Perforation Process: Numerical-Model Buildup and Parametric Study," **SPE Journal** **24(05)**, 2097-2110; **Society of Petroleum Engineers** (10.2118/183836-pa)

A. Ben Slimane; A. Najjar; R.T. ElAfandy; D.P. San-Roman-Alerigi; D.H. Anjum; T.K. Ng; B.S. Ooi, "On the phenomenon of large photoluminescence red shift in GaN nanoparticles," **Nanoscale Research Letters** **8(1)**, 342; **Springer** (10.1186/1556-276x-8-342)

D.P. San-Roman-Alerigi; A. Ben Slimane; T.K. Ng; M. Alsunaidi; B.S. Ooi, "A possible approach on optical analogues of gravitational attractors," **Optics Express** **21(7)**, 8298-8310; **Optica (OSA)** (10.1364/oe.21.008298) **FIRST AUTHOR · FEATURED, LASER FOCUS WORLD**

D.P. San-Roman-Alerigi; D.H. Anjum; Y. Zhang; X. Yang; A. Ben Slimane; T.K. Ng; M.N. Hedhili; M. Alsunaidi; B.S. Ooi, "Electron irradiation induced reduction of the permittivity in chalcogenide glass (As<sub>2</sub>S<sub>3</sub>) thin film," **Journal of Applied Physics** **113(4)**, 044116; **AIP Publishing** (10.1063/1.4789602) **FIRST AUTHOR**

D.P. San-Roman-Alerigi; T.K. Ng; M. Alsunaidi; Y. Zhang; B.S. Ooi, "Generation of J<sub>0</sub>-Bessel-Gauss beam by an heterogeneous refractive index map," **Journal of the Optical Society of America A** **29(7)**, 1252-1258; **Optica (OSA)** (10.1364/josaa.29.001252) **FIRST AUTHOR**

## Industry / technical journal articles (4) — Aramco Journal of Technology

S.I. Batarseh; D.P. San-Roman-Alerigi; S. Marshall; R.M. Graves, "First Industrial High Power Laser Perforation Deployment," **Aramco Journal of Technology, Winter 2023, p.2 (lead article); Saudi Arabian Oil Company (ISSN 1319-2388)** [LEAD ARTICLE](#)

W.J. Assiri; D.P. San-Roman-Alerigi; S.I. Batarseh, "Effect of High Power Laser on Mechanical Properties of Steel," **Aramco Journal of Technology, Spring 2023, p.82; Saudi Arabian Oil Company**

D.P. San-Roman-Alerigi; S.I. Batarseh; W.J. Assiri, "The Effect of High Power Laser on Organic-Rich Shales," **Aramco Journal of Technology, Fall 2021, p.2 (lead article); Saudi Arabian Oil Company** [FIRST AUTHOR · LEAD ARTICLE](#)

S.I. Batarseh; D.P. San-Roman-Alerigi; O.M. AlObaid; H.A. Othman, "Laser Gun: The Next Perforation Technology," **Aramco Journal of Technology, Summer 2019, p.2 (lead article); Saudi Arabian Oil Company** [LEAD ARTICLE](#)

## Preprints & Thesis

D.P. San Roman Alerigi; D.I. Ketcheson; B.S. Ooi, "A high-order finite volume method for Maxwell's equations in heterogeneous and time-varying media," arXiv:2307.11842 (2023) [FIRST AUTHOR](#)

D.P. San-Roman-Alerigi, "Controlling light propagation: a brief review," arXiv:1303.1014 (2013) [SOLE AUTHOR](#)

D.P. San Roman Alerigi, *Exploring Heterogeneous and Time-Varying Materials for Photonic Applications, Towards Solutions for the Manipulation and Confinement of Light*, King Abdullah University of Science and Technology (KAUST); advisor Prof. Boon S. Ooi (2014)

## CONFERENCE PAPERS — FULL RECORD (50)

S. Al Ghamdi; S. Batarseh; D. San Roman Alerigi; A. Alrashed; A. Al Harith, "Creating Sweet Spot in Tight and Unconventional Reservoirs Utilizing High Power Laser Technology," SPE-227837-MS, SPE Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain (2025)

A.A. Alrashed; S.I. Batarseh; D.P. San Roman Alerigi; A.M. Alharith, "Investigating the Role of Different Rock Properties in Laser Perforation," SPE-227625-MS, SPE Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain (2025)

A.A. Alrashed; S.I. Batarseh; D.P. San Roman Alerigi; T.Y. Alyousuf, "Laser Technology Accelerating the Path to Net Zero Upstream Operations," SPE-224865-MS, SPE Conference at Oman Petroleum & Energy Show, Muscat, Oman (2025)

H. Abdelmoula; R. Jabari; S. Batarseh; D. San Roman Alerigi, "Laser Descaling Robot for Long Range Scale Removal Applications," SPE-222608-MS, ADIPEC, Abu Dhabi (2024)

S.I. Batarseh; D.P. San Roman Alerigi; S. Marshall; S. BuHassan, "Validation of the First Industrial High-Power Laser Perforation Deployment," SPE-220900-MS, SPE Annual Technical Conference and Exhibition (ATCE), New Orleans, USA (2024)

W. Li; Y. Ma; D. San Roman Alerigi, "Automated Wellhead Monitoring Using Deep Learning from Multimodal Imaging," IPTC-23632-MS, International Petroleum Technology Conference (IPTC) (2024)

S. Alghamdi; S. Batarseh; A. Alharith; D.P. San Roman Alerigi; A.A. Alrashed, "Aligning Upstream Operations with Sustainable Technology: High Power Lasers," SPE GOTECH Conference (2024)

S.I. Batarseh; D.P. San Roman Alerigi; S. Marshall; R.M. Graves, "First Industrial High-Power Laser Perforation Deployment," SPE-215016-MS, SPE Annual Technical Conference and Exhibition (ATCE) (2023)

[KEY MILESTONE](#)

S.I. Batarseh; D.P. San Roman Alerigi; I. Zefzafy; H. Enezi, "Reducing Hydraulic Fracturing Breaking Pressure by Using High Power Lasers," SPE-215162-MS, SPE Annual Technical Conference and Exhibition (ATCE) (2023)

S.I. Batarseh; D.P. San Roman Alerigi; S. Marshall; K. Reduoane, "Successful Milestones Achieved Towards High-Power Laser Drilling," SPE-215166-MS, SPE Annual Technical Conference and Exhibition (ATCE) (2023)

A.A. Alrashed; J.L. Miskimins; S.I. Batarseh; D.P. San Roman Alerigi, "Advances in Modeling of High-Power Laser Perforation," SPE-216209-MS, ADIPEC, Abu Dhabi (2023)

S.I. Batarseh; D.P. San Roman Alerigi; A.A. Alrashed; I. Zefzafy, "Photonic-Hydraulic Fracturing Hybrid Approach Minimizing Breaking Pressure," SPE-216510-MS, ADIPEC, Abu Dhabi (2023)

D.P. San Roman Alerigi; W. Li; A. Cavazos Sepulveda; S. Batarseh, "AI inference of multiphysics properties using hyperspectral data," SPIE Optical Technology and Measurement for Industrial Applications Conference (2023) [FIRST AUTHOR](#)

A. Alharith; S. Batarseh; D. San Roman Alerigi, "High Power Laser Technology Toward the Downhole Applications: The Effect on Permeability," SPE-213948-MS, SPE Middle East Oil, Gas and Geosciences Show (MEOS GEO) (2023)

S. Batarseh; D.P. San Roman Alerigi; A. Al Harith, "Key Success Factor for High Power Laser Deployment: Strategy and Execution," SPE-213438-MS, SPE Middle East Oil, Gas and Geosciences Show (MEOS GEO) (2023)

S. Batarseh; S. Al Mutairi; M. Alqahtani; W. Assiri; D. San Roman Alerigi; S. Marshal, "First Industrial Flowlines Descaling Field Deployment Utilizing High Power Laser Technology," SPE-209972-MS, SPE Annual Technical Conference and Exhibition (ATCE) (2022)

D.P. San Roman Alerigi; S. Mutairi; S. Batarseh; W. Assiri, "Principles and Advantages of High-Power Lasers for Descaling Surface Equipment," SPE-209977-MS, SPE Annual Technical Conference and Exhibition (ATCE) (2022) **FIRST AUTHOR**

S. Batarseh; S. Mutairi; D. San Roman Alerigi; A. Al Harith, "Laser Technology for Downhole Applications; Past, Present and Future," SPE-204661-MS, SPE Middle East Oil & Gas Show and Conference (MEOS) (2021)

D. San Roman Alerigi; S. Batarseh; W. Assiri, "The Effect of High Power Laser on Organic-Rich Shales," SPE-204809-MS, SPE Middle East Oil & Gas Show and Conference (MEOS) (2021) **FIRST AUTHOR**

S. Batarseh; D. San Roman Alerigi; A. Al Harith; W. Assiri, "Thermal Effect on Formation Stability Due to Heterogeneity," SPE-204663-MS, SPE Middle East Oil & Gas Show and Conference (MEOS) (2021)

S. Batarseh; W. Assiri; D. San Roman Alerigi; B. Khaldi, "Formation Heterogeneity Effect on Steam Injection and Propagation," SPE-207970-MS, ADIPEC, Abu Dhabi (2021)

D. San Roman Alerigi; J.O. Alvarez; S.I. Batarseh; B. Al-Khaldi, "Integrated Multi-Frequency Characterization of Sandstone Rocks," SPE-202920-MS, ADIPEC, Abu Dhabi (2020) **FIRST AUTHOR**

A.M. Al-Harith; S.I. Batarseh; D. San Roman Alerigi; W. Asiri, "Overview of Recent Waterless Stimulation Technologies," SPE-202762-MS, ADIPEC, Abu Dhabi (2020)

D. San Roman Alerigi; S.I. Batarseh; W. Assiri, "Advances Towards a Comprehensive Model of Laser-Rock Interaction," OTC-30196-MS, Offshore Technology Conference Asia (OTC Asia), Kuala Lumpur (2020) **FIRST AUTHOR**

H. Othman; S.I. Batarseh; D. San Roman Alerigi; A. Alhubai, "Effective Wellbore Stimulation Methods Utilizing Advanced Thermal Technologies," OTC-30056-MS, Offshore Technology Conference Asia (OTC Asia) (2020)

S.I. Batarseh; D. San Roman Alerigi; A.M. Al-Harith; W. Assiri, "Effects of Rocks Heterogeneity on Thermal Propagation: An Integrated Measurement and Analysis Approach," OTC-30130-MS, Offshore Technology Conference Asia (OTC Asia) (2020)

S.I. Batarseh; A.M. Al-Harith; W. Assiri; D. San Roman Alerigi, "Laser Drilling in Extremely Hard Formation," OTC-30416-MS, Offshore Technology Conference Asia (OTC Asia) (2020)

W. Assiri; A. Asiri; D. San Roman-Alerigi; S.I. Batarseh; M. Salman, "A New Horizon in Corrosion Prevention Using State of the Art High Power Laser Cladding Technology," SPE-201617-MS, SPE Annual Technical Conference and Exhibition (ATCE) (2020)

H. Othman; D. San Roman Alerigi; S.I. Batarseh; O. Al Obaid, "Efficiency of High Power Laser in Carbonate Formations," SPE-197480-MS, ADIPEC, Abu Dhabi (2019)

S.I. Batarseh; D. San Roman Alerigi; A. Al Harith; H. Othman, "Laser Perforation: The Smart Completion," SPE-197192-MS, ADIPEC, Abu Dhabi (2019)

S.I. Batarseh; D. San Roman Alerigi; O. Al Obaid; H. Othman, "Laser Gun: The Next Perforation Technology," SPE-194775-MS, SPE Middle East Oil and Gas Show and Conference (MEOS) (2019)

D.P. San-Roman-Alerigi; S. Batarseh; W. Li; H. Othman, "Machine Learning and the Analysis of High-Power Electromagnetic Interaction with Subsurface Matter," SPE-195118-MS, SPE Middle East Oil and Gas Show and Conference (MEOS) (2019) **FIRST AUTHOR**

D.P. San-Roman-Alerigi, "The Role of Transient Perturbations and Heterogeneities in Subsurface Wave Propagation - A Scalable Numerical Solution," SPE-194888-MS, SPE Middle East Oil and Gas Show and Conference (MEOS) (2019) **SOLE AUTHOR**

S.I. Batarseh; D. San Roman Alerigi; J. Reece; H. Othman, "Downhole High-Power Laser Tools Development and Evolutions," SPE-193064-MS, ADIPEC, Abu Dhabi (2018)

D.P. San-Roman-Alerigi; Y. Han; H. Othman; S.I. Batarseh, "Geomechanical and Thermal Dynamics of Distributed and Far-Field Dielectric Heating of Rocks Assisted by Nano-Enablers - A Numerical Exploration," SPE-188790-MS, ADIPEC, Abu Dhabi (2017) **FIRST AUTHOR**

S.I. Batarseh; R.M. Graves; D.P. San-Roman-Alerigi; K. Chand, "Laser Perforation: Lab to the Field," SPE-188729-MS, ADIPEC, Abu Dhabi (2017)

A.M. Gomaa; D.P. San-Roman-Alerigi; K. Alnoaimi; A.J. Moraes, "Parameters Affecting the Fractured Well Productivity: Proppant Pack vs. Pillar Fracture - Numerical Study," SPE-188820-MS, ADIPEC, Abu Dhabi (2017)

A.M. Gomaa; D.P. San-Roman-Alerigi; K. Alnoaimi; G.A. Al-Muntasheri, "Fractured Well Productivity: Proppant Pack vs. Pillar Fracture - Numerical Study," SPE-188127-MS, SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition (2017)

S.I. Batarseh; H. Othman; D.P. San-Roman-Alerigi; S. Kumaoka, "Microwave With Assisted Ceramic Materials to Maximize Heat Penetration and Improve Recovery Efficiency of Heavy Oil Reservoirs," SPE-183773-MS, SPE Middle East Oil & Gas Show and Conference (MEOS) (2017)

Y. Han; Y. Fang; D.P. San-Roman-Alerigi; S.I. Batarseh, "Numerical Modeling of Thermal-Mechanical Interaction Process in High Power Electromagnetic Heating of Rocks," SPE-183836-MS, SPE Middle East Oil & Gas Show and Conference (MEOS) (2017)

D.P. San-Roman-Alerigi; Y. Han; S.I. Batarseh, "Thermal and Geomechanical Dynamics of High Power Electromagnetic Heating of Rocks," SPE-183917-MS, SPE Middle East Oil & Gas Show and Conference (MEOS) (2017) **FIRST AUTHOR**

D.P. San-Roman-Alerigi; S.I. Batareseh; Y. Han, "Numerical Modeling of Thermal and Mechanical Effects in Laser-Rock Interaction - An Overview," ARMA-2016-142, 50th US Rock Mechanics / Geomechanics Symposium (ARMA), pp. 1946-1956 (2016)

FIRST AUTHOR

D.P. San Roman Alerigi, "Heterogeneous, nonlinear and time varying materials: from traveling Bragg mirrors to spontaneous ring resonators and multiphase flow characterization," SPIE Photonics Europe (2016)

SOLE AUTHOR

A.A. Al-Jabr; D.P. San Roman Alerigi; B.S. Ooi; M.A. Alsunaidi, "An FDTD algorithm for simulating light propagation in anisotropic dynamic gain media," Proc. SPIE 9134, 91341U; SPIE Photonics Europe (2014)

D.P. San Roman Alerigi, "Revisiting Maxwell's equations: a wave propagation method for electromagnetics in spacetime-varying media," SPIE Photonics Europe (2014)

SOLE AUTHOR

A. Ben Slimane; A. Najar; T.K. Ng; D.P. San-Roman-Alerigi; D.H. Anjum; B.S. Ooi; C. Shen, "Surface States Effect on the Large Photoluminescence Redshift in GaN Nanostructures," Asia Communications and Photonics Conference (ACP) 2013 (2013)

D.P. San-Roman-Alerigi; T.K. Ng; B.S. Ooi; M. Alsunaidi, "Photonic analogies of gravitational attractors," 2013 Saudi International Electronics, Communications and Photonics Conference (SIECPC), IEEE (2013)

FIRST AUTHOR

D.P. San-Roman-Alerigi; A. Ben Slimane; T.K. Ng; M. Alsunaidi; B.S. Ooi, "On a pragmatic approach to optical analogues of gravitational attractors," arXiv:1212.1990, IEEE Photonics Global Conference (PGC) 2012 (2012)

FIRST AUTHOR

D.P. San Roman Alerigi; T.K. Ng; Y. Zhang; A. Ben Slimane; B.S. Ooi, "Monolithic Silicon-based Gauss to J0 Bessel beam converter," Progress In Electromagnetics Research Symposium (PIERS) 2012 (2012)

H. Al-Falih; Y. Khan; Y. Zhang; D.P. San-Roman-Alerigi; D. Cha; B.S. Ooi; T.K. Ng, "Fabrication of tuning-fork based AFM and STM tungsten probe," 8th Intl. Conf. on High-Capacity Optical Networks and Emerging Technologies (HONET 2011), pp. 190-192 (2011)

## PATENTS — FULL RECORD

### Granted (26)

**US 12669403 B2** + Sibling applications: US 2026/0029227 A1 (speckle reflectometry); US 2026/0028895 A1 (shadowgraphy) — Measuring wellhead displacement using beam deflection (2026)

**US 12590836 B2** + Pre-grant publication US 2024/0377249 A1 — Optical energy apparatus, systems, and methods (2026)

SOLE INVENTOR

**US 12559366 B2** + Pre-grant publication US 2024/0327211 A1 — Photocatalytic conversion of hydrogen sulfide to hydrogen (2026)

LEAD INVENTOR

**US 12682216 B2** + Pre-grant publication US 2024/0361264 A1 — Method and system for predicting properties of rock samples using thermal sensing and machine learning (2025)

SOLE INVENTOR

**US 12487075 B2** + Pre-grant US 2025/0327656 A1; sibling application US 2025/0327399 A1 — Characterizing rock properties with optical energy sources in a wellbore (2025)

**US 12459816 B2** + Pre-grant US 2024/0059564 A1; continuation US 2026/0035245 A1 — Hydrogen sulfide dissociation using reactive sulfur species (2025)

**US 12430795 B2** + Pre-grant US 2024/0331190 A1 — Automated wellhead monitoring using imaging and computer vision (2025)

**US 12394183 B2** + Pre-grant US 2024/0177457 A1 — Generating training samples via 3D modeling for greenhouse gas emission detection (2025)

**US 12359554 B2** + Pre-grant US 2022/0290553 A1 — Real-time multimodal radiometry for subsurface characterization during high-power laser operations (2025)

**US 12253459 B2** + Pre-grant US 2023/0100258 A1 — In-situ monitoring of chemical fingerprints in oilfield applications (2025)

**US 12181418 B2** + Pre-grant US 2024/0142383 A1 — System and method for in-situ probing of a conduit (2024)

**US 12181406 B2** + Pre-grant US 2024/0201079 A1 — Contactless measurement of rock wettability by photonic techniques (2024)

**US 12140850 B2** + EP 4118301 B1; WO 2021/181144 A1; CN 115552096 B; SA 522440458 B1 (priority 2020-03-11) — Reconfigurable optics for beam transformation (2024)

LEAD INVENTOR · 4-JURISDICTION FAMILY

**US 12029125 B2** + Related family member US 11611029 — Methods to harvest thermal energy during subsurface high power laser transmission (2024)

**US 11905778 B2** — Downhole laser tool and methods (2024)

**US 11867629 B2** — 4D chemical fingerprint well monitoring (2024)

**US 11852005 B2** — Deformation monitoring mechanism with multi-pixel angle-sensitive laser ranging (2023)

**US 11725504 B2** + WO 2022/251833 A1; CA 3220011 A1; EP 4348176 A1; JP 2024-520436 A. Pre-grant US 2022/0372868 A1 — Contactless real-time 3D mapping of surface equipment (2023)

LEAD INVENTOR · 5-JURISDICTION FAMILY

**US 11619097 B2** — System and method for laser downhole extended sensing (2023)

**US 11604323 B2** — Methods to harvest electromagnetic energy during subsurface high power laser transmission (2023)

**US 11585976 B2** + EP 4067735 A1; SA 122430881 B1 (granted 2024-06-09). Pre-grant US 2022/0317365 A1. No WO/CN/JP/CA member. — Optical fiber with corrugations (2023)

WITH UNIVERSITY OF SOUTHAMPTON

**US 11555958 B2** + EP 4067952 A2 / EP 4067952 B1 (granted approx. 2024-12-11). Pre-grant US 2022/0317369 A1. — Nested anti-resonant nodeless optical fiber and subsurface system using the same (2023)

WITH UNIVERSITY OF SOUTHAMPTON

**US 11378503 B2** — Numerical modeling of laser perforating process (2022)

**US 10738579 B2** + US 10731450 B2; MA 52686 A (Morocco); EP 3797206; WO 2020/021321 — Methods of using a laser-induced plasma tool (2020)

**US 10731450 B2** + US 10738579 B2; MA 52686 A (Morocco); EP 3797206; WO 2020/021321. Pre-grant US 2020/0032635 A1 — Laser-induced plasma tool (2020)

**US 10365537 B1** — Directional sensitive fiber optic cable wellbore system (2019)

### Pending applications (11)

**US 2026/0029227 A1** + Sibling of US 12669403 B2 (beam deflection) and US 2026/0028895 A1 (shadowgraphy) — Measuring wellhead displacement using speckle reflectometry

**US 2026/0028895 A1** + Sibling of US 12669403 B2; US 2026/0029227 A1 — Measuring wellhead displacement using shadowgraphy

**US 2026/0035245 A1** + Continuation of US 12459816 B2 — Hydrogen sulfide dissociation using reactive sulfur species (continuation)

**US 2025/0368507 A1** — Steam-methane reforming in hydrogen production

**US 2025/0327399 A1** + Sibling of US 12487075 B2 — Determining wellbore displacement with optical energy sources in a wellbore

**US 2025/0214869 A1** — Processing a fluid emulsion

**US 2024/0309705 A1** — Downhole high-power laser tool for subsurface applications

**US 2024/0141777 A1** — Distributed fiber sensing in a packer for permanent casing and formation deformation monitoring

**US 2024/0084695 A1** — Casing deformation monitoring

**US 2024/0060411 A1** — Photolytic conversion of hydrogen sulfide in a gas mixture produced by a hydrocarbon producing well and/or a gas treatment unit

**US 2022/0351037 A1** — Method and system for spectroscopic prediction of subsurface properties using machine learning

All patents assigned to Saudi Arabian Oil Company, except the hollow-core-fibre family (co-assigned with the University of Southampton). Jurisdiction families are consolidated to a single entry; component numbers (EP/WO/CA/CN/SA/JP/MA) are noted where applicable.

## AWARDS & HONORS

- **2026** — Finalist, Best Pipeline Rehabilitation Technology (Gulf Energy Information Excellence Awards (e2 Awards)): High-Power Laser Descaler Technology for Surface Flowlines — Team/technology recognition; not an individual award.
- **2026** — Finalist, Best Controls, Instrumentation & Automation Technology – Upstream (Gulf Energy Information Excellence Awards (e2 Awards)): WATCH: Wellsite Asset Tracking for Control of Hazards — Team/technology recognition; not an individual award.
- **2022** — Winner, CEO Excellence Award (Saudi Aramco)
- **2010** — Winner, Academic Excellence Award (King Abdullah University of Science and Technology)
- **2009** — Winner, Provost Award (King Abdullah University of Science and Technology)
- **2008** — Winner, Discovery Scholar (King Abdullah University of Science and Technology)
- **2008** — Winner, Research Scholarship (Caixa Catalunya – Institut de Ciències Fotòniques (ICFO))
- **2008** — Winner, PAPIIME Research Scholarship (UNAM)
- **2007** — Winner, REU Research Scholarship (University of Maryland)

## MEDIA & CORPORATE RECOGNITION

- **2023** — Saudi Aramco Laser Research Team recognition by SVP Waleed A. Al-Mulhim (The Arabian Sun / Aramco Life): Recognition for the first industrial high-power laser perforation field trial.
- **2013** — Optics Express paper featured in trade press (Laser Focus World (July 2013)): Coverage of 'A possible approach on optical analogues of gravitational attractors' (Opt. Express 21, 8298).

## PROFESSIONAL AFFILIATIONS & LEADERSHIP

- Member — IEEE (2010) · American Physical Society (2011) · Optica (OSA) (2012) · SIAM (2012) · Sociedad Mexicana de Física (2012) · The Competitiveness Institute (2012)

- **2009** — Founding Scholar & Founding Chair, KAUST Graduate Student Council: King Abdullah University of Science and Technology
- **2012–2013** — Vice-President, KAUST Graduate Student Council: King Abdullah University of Science and Technology — Governance liaison between administration, staff and the student body.
- **2013–** — Co-founder & lead developer, Maxwell-GEMS: Open numerical toolbox for electromagnetic wave propagation (FEM, FDTD, spectral, paraxial methods)

## ADDITIONAL SKILLS & CERTIFICATIONS

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**Languages:** Spanish (native), English (fluent), French (working).

**Technical certifications:** Laser Safety; Laboratory Safety; Clean-room Laboratory Safety. Trained and certified user of Scanning Electron Microscopy (SEM), Focused Ion Beam (SEM-FIB), Pulsed Laser Deposition (PLD), electron-beam deposition, and Raman/photoluminescence spectroscopy.

**Software & languages:** MATLAB, Python, Fortran, C, LaTeX, PETSc; COMSOL, Mathematica, RSoft, OriginLab; FEM, FDTD, Spectral / paraxial methods, Inverse-problem design.